

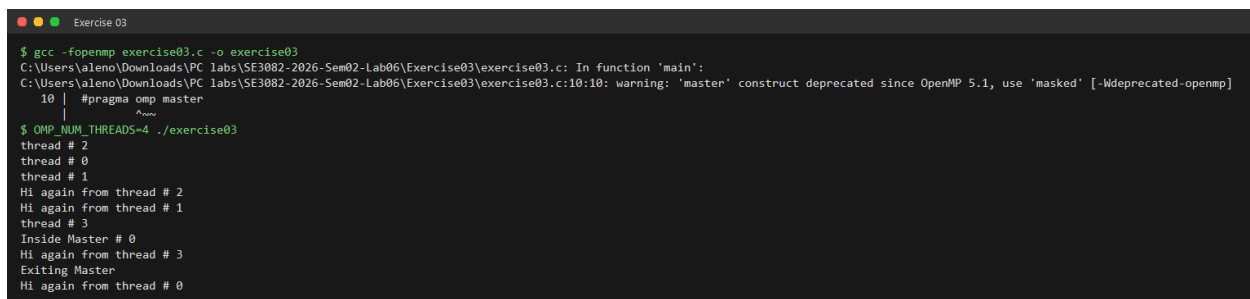
Exercise 3 - The #pragma omp master directive

Source file : exercise03.c

Build and execute

```
gcc -fopenmp exercise03.c -o exercise03
```

```
OMP_NUM_THREADS=4 ./exercise03
```



```
Exercise 03
$ gcc -fopenmp exercise03.c -o exercise03
C:\Users\aleno\Downloads\PC labs\SE3082-2026-Sem02-Lab06\Exercise03\exercise03.c: In function 'main':
C:\Users\aleno\Downloads\PC labs\SE3082-2026-Sem02-Lab06\Exercise03\exercise03.c:10:10: warning: 'master' construct deprecated since OpenMP 5.1, use 'masked' [-Wdeprecated-openmp]
   10 | #pragma omp master
      | ~~~~~
$ OMP_NUM_THREADS=4 ./exercise03
thread # 2
thread # 0
thread # 1
Hi again from thread # 2
Hi again from thread # 1
thread # 3
Inside Master # 0
Hi again from thread # 3
Exiting Master
Hi again from thread # 0
```

What the master directive does in this program

- * Only the master thread (thread id 0) ever enters the master block, so the line "Inside Master # 0" is printed exactly once and always by thread 0.
- * In contrast to single, master carries no implicit barrier: the rest of the team simply carries on. This is visible in the output, where thread 3 has already printed "Hi again" before the master block runs, and the master lines end up mixed in with the other threads' output.
- * gcc reports that master has been deprecated since OpenMP 5.1; its replacement, #pragma omp masked, behaves in exactly the same way.